How Do We Actually Know Anything?

A friendly tour of the philosophy of knowledge —
no background needed, real examples included

Featuring: swans, planets that don't exist, the sun (probably) rising tomorrow, and a toolkit for thinking more clearly

1. The Starting Puzzle: How Do You Know the Sun Will Rise Tomorrow?

Take a belief you're completely sure of: **the sun will rise tomorrow.** Now ask: how do you actually know that?

Your honest answer is probably: "because it has risen every single day so far." This way of thinking has a name: **induction**. It means looking at many past examples and jumping to a general rule that also covers cases you have *not* seen yet.

Everyday examples of induction:

- You've seen thousands of white swans → "all swans are white."
- Every glass you've ever dropped has fallen → "dropped things always fall."
- Your friend has always been on time → "she'll be on time tomorrow."

Notice the little **leap** in each one: the conclusion says more than the evidence actually guarantees. You haven't seen *all* swans. You haven't lived tomorrow yet. That small gap turns out to be one of the biggest problems in all of philosophy.

Hume's problem: the leap can't be justified

The Scottish philosopher **David Hume** (1700s) asked a devastating question: *what justifies the leap from "it always happened before" to "it will happen again"?*

The only answer anyone can give is: "well, trusting the past has always worked before." But look closely — that answer is itself an example of the very leap it's trying to justify! It's circular, like proving you're honest by saying "trust me, I'm honest."

Hume's conclusion: we can never logically *prove* that the future will resemble the past. Our deepest expectations about the world rest on **habit**, not on proof.

And history delivered the perfect symbol: Europeans were certain all swans were white — millions of confirmations — until they reached Australia and found **black swans**. One single counterexample destroyed a rule built on a mountain of evidence.

"But physics proves it!" — nice try

A natural objection: "we don't just *assume* the sun rises — we know the physics! Gravity explains it!" But ask: how do we know the law of gravity? By observing regularities and generalizing them... which is induction again, just one level up. Asking "why will gravity hold tomorrow?" gets the answer "because it always has" — the same circle.

Even worse for the objection: the physics we were once *most certain about* turned out to be wrong. For 200 years everyone "knew" Newton's theory of gravity was exact. Then the planet Mercury's orbit didn't quite fit, and Einstein replaced Newton's whole picture. Deep explanations are wonderful — but depth is not the same as certainty.

2. Popper's Clever Flip: Stop Proving, Start Attacking

The philosopher **Karl Popper** (1900s) accepted Hume's point — you can't prove a general rule by piling up examples — and then flipped the whole game around.

The asymmetry:

• No number of white swans can ever **prove** "all swans are white."

• But **one** black swan **disproves** it instantly.

Confirming is weak and never final. Refuting is powerful and decisive.

So Popper said: good thinking isn't about collecting confirmations of what you already believe. It's about making bold, precise claims that *could* be proven wrong — and then genuinely trying to break them. This is called **falsification**.

A theory that no possible observation could ever contradict ("everything that happens confirms my horoscope!") isn't deep — it's empty. If nothing can count against an idea, the idea isn't really saying anything.

A one-line test you can use anywhere

When someone (including you!) states a strong belief, ask: **"What would have to happen for you to admit this is wrong?"**

If the answer is "nothing could ever show it's wrong" — be suspicious. That belief is protected from reality, not supported by it.

3. But Wait — What Counts as Disproof? (It Gets Messy)

Here's the sharp objection that came up in our conversation: *what if a test fails, but I just blame something else? What if what counts as "disproof" depends on my own beliefs?* That objection turns out to be exactly right, and it's what philosophers spent the rest of the century wrestling with.

You never test one idea alone

Simple example: you believe "water boils at 100°C." You test it. The thermometer reads 90°C. Is your belief refuted? Not necessarily! Maybe the thermometer is broken. Maybe you're up a mountain (water boils cooler at altitude). Without noticing, you were also relying on "the thermometer works" and "I'm at sea level."

When the test fails, logic only tells you that **something in the whole bundle** is wrong. It never points a finger at which part. This is called the **Duhem–Quine problem** (after the two philosophers who spelled it out).

The best story in science: Neptune vs. Vulcan

Two nearly identical situations, with opposite endings:

Uranus (happy ending): the planet Uranus wasn't moving the way Newton's theory predicted. Instead of declaring Newton wrong, astronomers blamed a hidden assumption: "maybe there's an unseen planet pulling on it." They calculated where it should be, pointed their telescopes... and **discovered Neptune**. Theory saved, new planet found!

Mercury (plot twist): Mercury's orbit was *also* off. Astronomers tried the exact same trick — "there must be a hidden planet, let's call it Vulcan!" They searched for decades. **Vulcan doesn't exist.** This time, Newton's theory itself was wrong, and it took Einstein to fix it.

Same failure, same rescue attempt — one was brilliant, one was a dead end. And crucially, **no rule of logic could tell them in advance which case they were in.** This is why "just falsify it" is too simple: you can always rescue a cherished belief by blaming something else. Nothing in pure logic stops you.

4. Two Answers to the Mess: Kuhn and Lakatos

Kuhn: "your framework decides what counts"

Thomas Kuhn studied how science actually happens and noticed: scientists mostly work *inside* a big shared framework of assumptions — he called it a **paradigm** — and treat weird results as puzzles to solve later, not as reasons to panic.

Only when problems pile up for years *and* a better framework appears does the community switch — a **paradigm shift** (like Newton → Einstein). Kuhn's unsettling claim: what even *counts* as evidence depends on the paradigm you already hold. Before Copernicus, "the sun moves across the sky" was an obvious fact proving Earth sits still. After him, the *same* sky proves Earth spins. Same window, different meaning.

Lakatos: "you may protect your beliefs — if they keep earning it"

Imre Lakatos agreed that people protect their core ideas, but refused to accept that belief-change is just fashion. His picture: a big theory has a **hard core** (the central ideas you won't give up) surrounded by a **protective belt** (adjustable side-assumptions you're allowed to tweak when things go wrong).

His test for honest vs. dishonest protecting:

> **Progressing ✓** — your adjustments predict **new things that turn out true**. ("There's a hidden planet" → Neptune found. The patch produced a discovery!)
>
> **Degenerating ✗** — your adjustments only **explain away failures after the fact** and never predict anything new. ("Vulcan must be there... somewhere... keep looking...")

So Lakatos' rule of thumb: defending your beliefs is fine *while the defence keeps generating fresh, successful predictions*. The moment your defending becomes pure damage control, that's your rational signal to let go. Judge ideas by their **track record over time**, not by a single win or loss.

Quine: the web of belief

W.V.O. Quine zoomed all the way out: picture *everything you believe* as one connected web. At the centre sit things you'd almost never give up (logic, math, "objects exist"). At the edges sit small everyday beliefs ("the thermometer read 90°"). When reality clashes with the web, you choose where to make the repair — and usually it's smarter to fix an edge than rip out the centre.

Lakatos' picture is basically a slice of Quine's web *plus a rulebook*: Quine describes how beliefs hang together; Lakatos adds the test for when protecting the centre is wise and when it's cheating.

5. "So Nothing Is True and We're Just Guessing?" — No!

This is where the conversation got personal: if nothing can be proven, doesn't that mean we never know anything and everything is a guess? Three ideas rescue us.

Rescue #1: the world can be solid even if proof isn't

Keep two things separate: **how the world is** versus **what we can prove about it**. Hume only attacked the second. Gravity may hold perfectly, everywhere, forever — the world itself may be completely regular. What we lack is an airtight *proof* of that. "I can't prove X" is not the same as "X isn't real." The uncertainty lives in our knowledge, not necessarily in the world.

Rescue #2: "not proven" does not mean "all guesses are equal"

Compare two claims that are both, technically, "unproven":

- "The sun will rise tomorrow" — survived billions of tests, woven into physics that predicts eclipses **to the second**.
- "The sun will turn into a dragon tomorrow" — predicts nothing, connects to nothing, contradicts everything.

If your worldview calls both of these "guesses," it has erased the most important distinction a mind can make. The real scale was never *proven vs. guessed* — it's **weakly-tested vs. heavily-tested belief**, and that scale is fully intact. You already live by it: you board planes, take medicine, cross bridges — all "unproven," all rationally trusted because they've survived enormous testing.

Rescue #3: reality pushes back

Here's the beautiful asymmetry: the world never hands you a certificate saying "this belief is TRUE" — but it constantly **corrects** you. The glass shatters. The recipe fails. Vulcan isn't there. You're not guessing into a void; you're in a feedback loop with something that pushes back. A belief that has survived years of things that *could have killed it* is a completely different creature from a fresh guess.

> ***We don't get proof — we get correction. And a belief that has survived heavy correction is the most reliable thing a mind can hold.***

What about math? Isn't 2+2=4 just true?

Yes — math and logic are the one place you get genuine proof. Denying "2+2=4" isn't unlikely, it's a contradiction. But the certainty is **conditional**: *given* the starting rules (axioms), the conclusions follow with certainty. What math can't guarantee is that its rules **describe the real world**.

Famous example: for 2,000 years, Euclid's geometry ("triangle angles add to 180°") seemed like eternal truth about space itself. Then physicists discovered that near massive objects, real space *bends* — and triangle angles *don't* add to 180°. Euclid was never "wrong" — his theorems still follow perfectly from his axioms. But whether the universe actually follows those axioms was a question only observation could answer. Perfect proof inside the map; no guarantee the map matches the territory.

And probability? Can't we be 99.999% sure?

Yes — and that's exactly the right move! When a theory predicts eclipses to the second, a false theory would almost never get that lucky, so your confidence can rationally climb to 99.999%. Acting on such near-certainty isn't sloppy — it's wisdom. But two honest footnotes: (1) that number measures *your confidence*, not a stamp on the world; (2) precise predictions can sit on top of a wrong *explanation*. Newton predicted the heavens beautifully for 200 years — and his story of *why* (a force pulling across space) turned out to be wrong. Great predictions, wrong "why." Keep those separate.

6. The Critical-Thinking Toolkit (Keep This Page)

Everything above compresses into habits you can actually use — in arguments, news, shopping decisions, health choices, everywhere.

1. Ask the Popper question.

"What would change my mind?" If you (or someone else) can't name *anything* that would count against a belief, the belief is being protected from reality. Decide **in advance** what would count as being wrong — because after the fact, we're all champions at explaining failures away.

2. Confirmation is cheap; survival is expensive.

A belief that has only ever been *confirmed* (by people looking for confirmation) is weak. A belief that has survived genuine attempts to break it is strong. Seek out the best argument *against* what you believe — not the dumbest version of it.

3. Remember the black swan.

"It's always been this way" is evidence, but never a guarantee. The more absolute a claim ("all," "never," "impossible"), the more a single counterexample can destroy it. Hold universal claims a little loosely.

4. When a test fails, remember the bundle.

A failed prediction doesn't tell you *which* assumption was wrong — the idea, the measurement, or a hidden side-assumption. Before declaring victory or defeat in an argument, ask: what else was being assumed here? (Thermometer broken? Or water really not boiling at 100°?)

5. Excuses are legal — but track their record.

Defending a belief by adjusting side-assumptions is fine (that's how Neptune was found!). The test is Lakatos': do the adjustments **predict new things that come true**, or do they only **explain away failures**? A belief that needs a fresh excuse every week is a Vulcan.

6. Repair the edge before the centre — but nothing is sacred.

When new information clashes with what you believe, it's usually rational to adjust small peripheral beliefs first (Quine). But "usually" isn't "always": sometimes the core itself is what's wrong, and clinging to it is the real mistake.

7. Don't confuse "unproven" with "just a guess."

Almost nothing outside math is provable — and yet beliefs differ enormously in how much testing they've survived. Vaccines and horoscopes are both "unproven" in the strict logical sense; they are not remotely equal. Ask "how hard has this been tested?", not "is this proven?"

8. Confidence is a dial, not a switch.

Replace "true/false" with "how confident, given the evidence?" It's fine — rational, even — to act decisively on 99.9% confidence. It's also fine to say "I'm 60% on this" and stay open. People who only have ON and OFF break when reality gets complicated.

9. Great predictions ≠ true explanation.

Someone (or some theory, or some influencer) being impressively right about *what* happens doesn't guarantee their story about *why* is correct. Newton predicted perfectly with a wrong "why" for two centuries. Trust track records, but hold explanations more loosely than predictions.

10. Welcome correction — it's the only teacher there is.

Since proof is off the menu, being corrected by reality (or by a good counter-argument) is not losing — it's the mechanism by which anyone ever gets less wrong. The goal isn't to be certain. It's to be the kind of thinker whose wrong ideas don't survive long.

7. One-Page Cheat Sheet

Idea	In one sentence	Remember it by
Induction	Generalizing from past examples to a rule about unseen cases.	White swans → "all swans are white"
Hume's problem	The leap from past to future can't be logically justified — only habit.	"It always worked before" is circular
Black swan	One counterexample kills a rule built on endless confirmations.	Australia's black swans
Popper / falsification	Don't try to prove ideas — try to break them; survivors earn trust.	"What would change my mind?"
Duhem–Quine	You always test a bundle of assumptions; failure doesn't say which is wrong.	Broken thermometer or boiling point?
Neptune vs. Vulcan	The same excuse can be a discovery or a dead end — logic can't tell in advance.	One planet existed, one didn't
Kuhn / paradigms	What counts as evidence depends on the framework a community already holds.	Same sky, before/after Copernicus
Lakatos	Protecting core beliefs is fine while it keeps predicting new true things.	Progressing vs. degenerating
Quine's web	Beliefs form a web: solid centre, adjustable edges; repair edges first.	Fix the edge, not the core (usually)
Math's certainty	Proof is real but conditional: certain inside the system, not about the world.	Euclid vs. curved space
Probability	Near-certainty is rational to act on — but it measures confidence, not truth.	99.999% ≠ 100%
Prediction ≠ explanation	Being right about what happens doesn't prove your story of why.	Newton: right sky, wrong "why"
The big consolation	We don't get proof — we get correction, and survivors of correction are gold.	Sun rising vs. sun-dragon

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The whole conversation in three lines: We can't prove our beliefs about the world — not even with physics, and math only proves things inside its own rules. But "unprovable" never meant "all equal": beliefs differ enormously in how much reality-testing they've survived, and reality corrects us constantly. So the goal of good thinking isn't certainty — it's building beliefs that survive honest attempts to break them, and letting go gracefully of the ones that don't.